

NASH BARGAINING PREFERENCE OPTIMIZATION

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ABSTRACT

Aligning one large language model (LLM) to several objectives requires two decisions: how to represent preferences within each objective and how to choose one compromise across objectives. The first decision is difficult because pairwise judgments can be cyclic in realistic settings. For a medical-advice prompt, annotators may prefer a concise answer to a verbose one, the verbose answer to an incomplete one, yet the incomplete answer to the concise one because it includes a critical safety caveat. No scalar ranking can satisfy all three comparisons. Nash Learning from Human Feedback avoids this restriction by treating pairwise preferences as game payoffs, but its multi-objective extensions still require an aggregation rule. We introduce Nash Bargaining Preference Optimization (NBPO), which evaluates each objective through a regularized preference game and applies Nash bargaining to its improvement over a common reference policy. Our main technical contribution is a practical optimizer for these policy-dependent game values: a constrained KL-proximal update handles zero surplus at the reference, a prompt-separable finite-pool solve accounts for adaptive opponents, and a pairwise log-ratio regression objective states the resulting policy target for an LLM without evaluating an intractable normalizer. Correcting the implementation makes that target realizable in an 8B policy: on prompts held out from training, the regression reaches Pearson 0.52 with 69% sign agreement, against roughly zero transfer before, though its remaining error is dominated by the magnitude of the induced update rather than its direction. Under exact population updates, the final iterate achieves convergence with rate $O(1/T)$. Controlled experiments show that adaptive game-valued objectives remain robust as preference nontransitivity increases. On a human-labeled SAFERLHF panel the trained policy then carries both objectives strictly above the reference on held-out prompts, closing most of the distance to the finite-pool optimum without losing instruction-following or mathematical accuracy — though an independent reward model ranks it below the reference on general prompts, so we report a compromise rather than a Pareto improvement.

1 INTRODUCTION

Reinforcement Learning from Human Feedback (RLHF) is a standard approach for aligning large language models (LLMs) with human judgments (Christiano et al., 2017; Ouyang et al., 2022). Standard RLHF learns a scalar reward from pairwise comparisons. Direct Preference Optimization removes the separately trained reward model, but still assumes that comparisons are generated by an implicit scalar reward through a Bradley–Terry (BT) model (Rafailov et al., 2023; Bradley & Terry, 1952). Both approaches therefore require every response to admit one scalar score whose ordering explains all pairwise preferences.

This assumption can fail even within a single alignment objective. Human judgments can reflect several latent attributes whose relative importance changes with the prompt and the pair being compared (Tversky, 1969; Wang et al., 2024). As an illustrative example, consider three answers to a medical-advice question. Response A is concise and actionable but omits an important caveat; response B is comprehensive but verbose; and response C is less complete but includes the critical safety warning. Annotators may prefer A to B for relevance, B to C for completeness, and C to A for safety and reliability. These comparisons are cyclic: $A \succ B$, $B \succ C$, and $C \succ A$. Fitting the first two comparisons with a scalar reward requires $r(A) > r(B) > r(C)$, which contradicts the third. As a result, optimizing the fitted scalar reward can favor a response that appears best under the learned

ranking but is still beaten by another response according to the original pairwise judgments (Munos et al., 2024).

Nash Learning from Human Feedback (NLHF) avoids this misspecification by treating prompt-conditioned comparisons directly as payoffs in a zero-sum preference game, and INPO, SPPO, and related methods provide scalable equilibrium-seeking updates (Munos et al., 2024; Zhang et al., 2025; Wu et al., 2025; Calandriello et al., 2024; Rosset et al., 2024; Zhou et al., 2025). Extending this view to multiple objectives introduces a second problem: after representing each objective without imposing transitivity, which compromise should one deployed policy realize? Scalar multi-objective methods expose this trade-off through weights, priorities, constraints, policy interpolation, or preference conditioning (Zhou et al., 2024; Ramé et al., 2023; Zhong et al., 2024b; Chakraborty et al., 2024; Son et al., 2026; Agnihotri et al., 2026). Recent game-theoretic methods preserve general pairwise feedback but combine objectives through Blackwell target sets or a worst-objective criterion (Blackwell, 1956; Bhatia et al., 2020; Zhang et al., 2026).

Classical Nash bargaining offers a priority-free way to select one compromise. It maximizes the product of each objective’s improvement over a common fallback (Nash, 1950). In alignment, the reference policy provides a natural fallback: an aligned policy should improve each objective relative to the model that would otherwise be deployed. The classical Nash solution is Pareto optimal and invariant to independent changes in the units used for different objectives. Prior alignment work has applied Nash welfare to learned scalar rewards, while game-based preference methods have focused mainly on minimax or Blackwell solutions (Zhong et al., 2024a; Navon et al., 2022; Zhang et al., 2026). What remains is a practical method for Nash bargaining when every objective value is itself defined by a policy-dependent preference game rather than by a fixed scalar reward.

We introduce Nash Bargaining Preference Optimization (NBPO) for this setting. Within each objective, NBPO evaluates the learner against an adaptive opponent that emphasizes responses exposing the learner’s weaknesses, while a KL penalty keeps the opponent near the reference policy. Across objectives, it maximizes the Nash product of the resulting improvements over that same reference. The classical bargaining properties matter here: at the population optimum no objective falls below the fallback, the outcome is Pareto efficient, and judge-specific units do not move the compromise.

The main technical challenge is optimization, not re-deriving the classical bargaining axioms. Three issues arise simultaneously: the reference policy has zero improvement and lies on the boundary of the logarithmic objective; each adaptive opponent depends on the policy being optimized; and an LLM cannot directly realize the normalized finite-action policy update. We address them with a constrained KL-proximal formulation, a low-dimensional dual whose weights are inversely proportional to achieved improvements, a prompt-separable direct solve of the finite-pool inner problem, and a partition-free pairwise log-ratio regression objective for projecting the finite-pool target into the language model.

Our contributions are:

- We formulate multi-objective alignment as Nash bargaining over objective-wise regularized preference-game values, separating within-objective preference representation from across-objective compromise selection.
- We derive a practical optimizer that handles the zero-surplus boundary and policy-dependent opponents. For each dual iterate, the sampled finite-pool problem separates into small strictly concave prompt-wise programs; their target policy is expressed for the LLM as a pairwise log-ratio regression objective, without fitting a scalar reward or evaluating a language-model partition function. With the implementation corrected, that regression transfers to prompts held out from training; we report its whole horizon trajectory and its remaining scale error rather than a single point.
- We prove $O(1/T)$ last-iterate convergence for exact population proximal updates. Feasibility-preserving controlled experiments isolate the benefit of adaptive game values under increasing nontransitivity, while a human-labeled SAFERLHF panel is reserved for testing Nash aggregation under natural objective conflict. Released natural datasets are not treated as evidence of within-objective cycles unless their comparison graphs identify such cycles.

2 BACKGROUND

General preference models. We use contextual-bandit notation. A prompt (context) x is drawn from a distribution $\mathcal{D}$, a response y is an action from a finite set $\mathcal{Y}_x$, and $\pi(y | x)$ is the policy. The reference policy is denoted by π_{ref} . For the finite-dimensional population analysis, the support of $\mathcal{D}$ and every $\mathcal{Y}_x$ are finite, and

$$\Pi = \prod_{x \in \text{supp}(\mathcal{D})} \Delta(\mathcal{Y}_x)$$

is the contextual policy simplex. Objective $k \in [K]$ is represented by a pairwise preference model $P_k(y \succ z | x) \in [0, 1]$ satisfying

$$P_k(y \succ z | x) + P_k(z \succ y | x) = 1, \quad P_k(y \succ y | x) = \frac{1}{2}. \quad (1)$$

We center each comparison at a tie,

$$\Delta_k(y, z | x) = P_k(y \succ z | x) - \frac{1}{2}, \quad \Delta_k(y, z | x) = -\Delta_k(z, y | x), \quad (2)$$

and define the expected objective- k payoff of learner policy π against comparator policy ν as

$$J_k(\pi, \nu) = \mathbb{E}_{x \sim \mathcal{D}, y \sim \pi(\cdot | x), z \sim \nu(\cdot | x)} \Delta_k(y, z | x). \quad (3)$$

Thus $J_k(\pi, \nu) > 0$ means that π is preferred to ν on average under objective k . This representation does not assume a scalar reward or transitive preferences.

Bargaining problems. A K -dimensional bargaining problem consists of a feasible value set $\mathcal{G} \subseteq \mathbb{R}^K$ and a disagreement point $d \in \mathcal{G}$, the outcome obtained if no agreement is reached. For a feasible value vector v , define its objective-wise surplus over disagreement as

$$s_k(v) := v_k - d_k, \quad s(v) = (s_1(v), \dots, s_K(v)). \quad (4)$$

A value vector is individually rational when $s(v) \geq 0$ componentwise, and strictly individually rational when $s(v) > 0$. On a compact, convex, and downward-closed feasible set with at least one strictly individually rational point, the classical Nash solution is

$$v^N \in \arg \max_{v \in \mathcal{G}: s(v) > 0} \sum_{k=1}^K \log s_k(v). \quad (5)$$

Equivalently, it maximizes the product of the objective-wise surpluses (Nash, 1950).

3 OBJECTIVE-WISE PREFERENCE GAMES

We first convert each objective's pairwise preference model into a scalar policy value without imposing a scalar reward on individual responses. For objective k , an adaptive comparator ν searches for responses against which the learner performs poorly. The reference policy π_{ref} anchors this comparator and later serves as the fallback in the bargaining problem.

Definition 1 (Entropy-regularized game value). *For objective k and temperature $\beta_k > 0$, define*

$$V_{k, \beta_k}(\pi) = \min_{\nu} \{J_k(\pi, \nu) + \beta_k \mathbb{E}_{x \sim \mathcal{D}} \text{KL}(\nu(\cdot | x) \| \pi_{\text{ref}}(\cdot | x))\}. \quad (6)$$

The comparator minimizes the learner's pairwise payoff, so it emphasizes responses that expose weaknesses of the current policy. The KL term prevents it from moving arbitrarily far from π_{ref} . Accordingly, a large β_k produces a local evaluation near the reference, whereas $\beta_k \downarrow 0$ approaches the unregularized worst-comparator value.

For a candidate comparator response z , let

$$m_k^\pi(z | x) := \mathbb{E}_{y \sim \pi(\cdot | x)} \Delta_k(y, z | x) \quad (7)$$

be the learner's expected margin against z . Smaller values identify stronger challenges. The inner problem in (6) has the closed-form solution

$$\nu_{k, \pi}^*(z | x) = \frac{\pi_{\text{ref}}(z | x) \exp\{-m_k^\pi(z | x)/\beta_k\}}{\mathbb{E}_{z' \sim \pi_{\text{ref}}(\cdot | x)} \exp\{-m_k^\pi(z' | x)/\beta_k\}}, \quad (8)$$

$$V_{k, \beta_k}(\pi) = -\beta_k \mathbb{E}_{x \sim \mathcal{D}} \log \mathbb{E}_{z \sim \pi_{\text{ref}}(\cdot | x)} \exp\{-m_k^\pi(z | x)/\beta_k\}. \quad (9)$$

Thus the opponent is obtained by reweighting reference responses toward those with smaller margins, and the game value is a soft minimum of these margins.

Proposition 1 (Game-value structure). *For every k and $\beta_k \geq 0$, V_{k,β_k} is continuous and concave in π . For $\beta_k > 0$, its policy-gradient coordinate is*

$$[\nabla_{\pi} V_{k,\beta_k}(\pi)]_{x,y} = \mathcal{D}(x) Q_k^{\pi}(y | x), \quad Q_k^{\pi}(y | x) := \mathbb{E}_{z \sim \nu_{k,\pi}^*(\cdot | x)} \Delta_k(y, z | x). \quad (10)$$

The proof is in Appendix A.1. The response-level score $Q_k^{\pi}(y | x)$ is the expected margin of response y against the current objective- k opponent; it therefore measures the local effect of increasing $\pi(y | x)$. An adaptive opponent can distinguish policies that have the same win rate against one fixed reference but different vulnerabilities to other responses. Appendix A.2 gives a finite cyclic example at the unregularized endpoint $\beta = 0$; this illustration is not used in the later assumptions or proofs.

4 NASH BARGAINING OVER GAME-VALUE IMPROVEMENTS

We now combine the objective-wise game values using the reference policy as the common fallback. For each objective,

$$d_k := V_{k,\beta_k}(\pi_{\text{ref}}), \quad s_k(\pi) := V_{k,\beta_k}(\pi) - d_k. \quad (11)$$

The surplus $s_k(\pi)$ is the improvement of π over the deployed reference under objective k .

Assumption 1 (Joint improvement and support). *The reference π_{ref} has full support, and there exists a policy $\bar{\pi}$ such that $s_k(\bar{\pi}) > 0$ for every k .*

The assumption states that at least one policy improves all objectives over the fallback. Under this condition, the NBPO population target is

$$\pi^N \in \arg \max_{\pi \in \Pi_+} F(\pi), \quad F(\pi) := \sum_{k=1}^K \log s_k(\pi), \quad \Pi_+ := \{\pi \in \Pi : s_k(\pi) > 0, \forall k\}. \quad (12)$$

Equivalently, introducing auxiliary lower bounds $u_k > 0$ yields

$$(\pi^N, u^N) \in \arg \max_{\pi, u \in \mathbb{R}_{++}^K} \sum_{k=1}^K \log u_k \quad \text{s.t.} \quad u_k \leq s_k(\pi), \quad k \in [K], \quad \pi \in \Pi. \quad (13)$$

Because the objective is increasing in every u_k , each constraint is tight at an optimum and $u_k^N = s_k(\pi^N)$.

To connect (12) to the classical bargaining problem, define

$$V_{\beta}(\pi) = (V_{1,\beta_1}(\pi), \dots, V_{K,\beta_K}(\pi)), \quad d = (d_1, \dots, d_K),$$

and the downward-closed feasible value set

$$\mathcal{G}_{\beta} := \{v \in \mathbb{R}^K : d \leq v \leq V_{\beta}(\pi) \text{ componentwise for some } \pi \in \Pi\}. \quad (14)$$

Proposition 2 (Policy-induced Nash bargaining). *Under Assumption 1, $\mathcal{G}_{\beta}$ is compact, convex, downward closed above d , and nondegenerate. The policies solving (12) realize the classical Nash solution of $(\mathcal{G}_{\beta}, d)$. Consequently, every optimizer has positive surplus in every objective and is Pareto optimal in game-value space; all optimizers induce the same game-value vector; and the optimizer set is unchanged by independent transformations*

$$(V_{k,\beta_k}, d_k) \mapsto (c_k V_{k,\beta_k} + b_k, c_k d_k + b_k), \quad c_k > 0.$$

Appendix A.3 verifies the feasible-set conditions. Once those conditions hold, the stated outcome properties are standard consequences of the classical Nash bargaining solution (Nash, 1950); we do not claim them as new bargaining axioms. In LLM alignment, they mean that the population solution improves every objective over the reference, cannot be Pareto dominated by another feasible policy, and does not change merely because different judges report their values in different numerical units. Appendix A.4 compares these guarantees with utilitarian and maxmin aggregation.

5 NASH BARGAINING PREFERENCE OPTIMIZATION

Optimizing (12) presents three practical difficulties. First, the reference has $s_k(\pi_{\text{ref}}) = 0$, so the logarithmic objective is undefined at initialization. Second, every game value contains an opponent that depends on the learner. Third, the population policy update must be realized by an autoregressive model. We handle these issues with a constrained proximal problem, a prompt-separable finite-pool inner solve, and neural log-ratio projection.

5.1 CONSTRAINED PROXIMAL UPDATE AND ADAPTIVE OBJECTIVE WEIGHTS

For $\pi \in \Pi_+$, Proposition 1 and the chain rule give

$$[\nabla_\pi F(\pi)]_{x,y} = \mathcal{D}(x) \sum_{k=1}^K \frac{Q_k^\pi(y | x)}{s_k(\pi)}. \quad (15)$$

The inverse-surplus factors emphasize objectives that have improved less. Because this expression cannot be evaluated at π_{ref} , outer stage t instead solves

$$\begin{aligned} (\pi_{t+1}, u_{t+1}) \in \arg \max_{\pi, u \in \mathbb{R}_{++}^K} \quad & \sum_{k=1}^K \log u_k - \frac{1}{\eta_t} D(\pi \| \pi_t) \\ \text{s.t.} \quad & u_k \leq s_k(\pi), \quad k \in [K], \\ & \pi \in \Pi, \end{aligned} \quad (16)$$

where $\pi_0 = \pi_{\text{ref}}$, $\eta_t > 0$, and $D(\pi \| \pi_t) = \mathbb{E}_{x \sim \mathcal{D}} \text{KL}(\pi(\cdot | x) \| \pi_t(\cdot | x))$.

The dual is

$$\min_{\lambda \in \mathbb{R}_{++}^K} \phi_t(\lambda), \quad \phi_t(\lambda) = - \sum_{k=1}^K \log \lambda_k - K + \max_{\pi \in \Pi} \left\{ \sum_{k=1}^K \lambda_k s_k(\pi) - \frac{1}{\eta_t} D(\pi \| \pi_t) \right\}, \quad (17)$$

where $\lambda = (\lambda_1, \dots, \lambda_K) \in \mathbb{R}_{++}^K$ is the vector of Lagrange multipliers. For fixed λ , let

$$\pi_t^*(\lambda) := \arg \max_{\pi \in \Pi} \left\{ \sum_{k=1}^K \lambda_k s_k(\pi) - \frac{1}{\eta_t} D(\pi \| \pi_t) \right\}. \quad (18)$$

The KL term makes the maximizing policy $\pi_t^*(\lambda)$ unique. Danskin’s theorem (Danskin, 1967) then allows us to differentiate the $\phi_t(\lambda)$ while holding this maximizing policy fixed, yielding

$$\nabla_k \phi_t(\lambda) = s_k(\pi_t^*(\lambda)) - \frac{1}{\lambda_k}. \quad (19)$$

The complete dual derivation is in Appendix A.5.

Proposition 3 (Inverse-surplus objective weights). *At an optimal primal–dual solution of (16),*

$$u_{t+1,k} = s_k(\pi_{t+1}), \quad \lambda_{t+1,k} = \frac{1}{u_{t+1,k}} = \frac{1}{s_k(\pi_{t+1})}. \quad (20)$$

Thus the constrained problem recovers the inverse-surplus weighting in (15) without selecting objective weights in advance.

5.2 DIRECT FINITE-POOL INNER SOLVE AND NEURAL PROJECTION

At one outer stage, sample learner candidates $\mathcal{Y}_x = \{y_{x,i}\}_{i=1}^{n_x}$ and comparator candidates $\mathcal{Z}_x = \{z_{x,j}\}_{j=1}^{m_x}$, and estimate the centered pairwise matrices $\hat{A}_{k,x}[i, j] = \hat{\Delta}_k(y_{x,i}, z_{x,j} | x)$. Let $p_{t,x} \in \Delta^{n_x}$ be the learner pool’s *occurrence measure* — the empirical importance weight each sampled candidate carries inside the finite program — and let μ_x be the corresponding occurrence measure on $\mathcal{Z}_x$. Both are counting measures over what was actually drawn, not the language model’s conditional law: with n_x i.i.d. draws from π_t and duplicates retained with multiplicity, $p_{t,x}(i) = 1/n_x$ regardless of the value of $\pi_t(y_{x,i} | x)$. The finite program below reweights that sampled support; the only bridge back to the model is the pool mapping $p_\theta(i) \propto p_{t,x}(i) \exp\{\log \pi_\theta(y_{x,i} | x) - \log \pi_t(y_{x,i} | x)\}$, which is what every finite-pool quantity we report is computed through. Reading $\text{softmax}_i \log \pi_\theta(y_{x,i} | x)$ as the pool distribution instead silently replaces the occurrence measure by a renormalized conditional and changes the sign of measured surpluses. For $p \in \Delta^{n_x}$, define

$$\hat{m}_{k,x}(p)_j := \sum_{i=1}^{n_x} p_i \hat{A}_{k,x}[i, j], \quad (21)$$

$$\hat{V}_{k,x}(p) := -\beta_k \log \sum_{j=1}^{m_x} \mu_{x,j} \exp\{-\hat{m}_{k,x}(p)_j / \beta_k\}. \quad (22)$$

The disagreement value $\widehat{d}_k$ is measured from an independently constructed reference–reference tensor, not set to zero.

For fixed λ , the empirical counterpart of (18) separates across prompts:

$$p_{t,x}^*(\lambda) \in \arg \max_{p \in \Delta^{n_x}} \left\{ \sum_{k=1}^K \lambda_k \widehat{V}_{k,x}(p) - \frac{1}{\eta_t} \text{KL}(p \| p_{t,x}) \right\}. \quad (23)$$

Each problem is strictly concave because of the KL term. We solve these small programs directly to a declared numerical tolerance and aggregate their values to obtain $\widehat{s}_k(p_t^*(\lambda))$. Dual stationarity is the K -dimensional condition $\widehat{s}_k(p_t^*(\lambda)) = 1/\lambda_k$, which we solve as a root problem in $\log \lambda$ over $\Lambda = [\lambda_{\min}, \lambda_{\max}]^K$; every reported run uses this solve and reports its projected KKT residual. A projected subgradient fallback on the same condition stalls three orders of magnitude short of that tolerance (Appendix A.5). In contrast, repeatedly alternating a closed-form opponent with the exponential policy map need not converge under strong nontransitivity; we retain that R -step map only as an ablation.

Once the dual converges, the finite-pool target defines the pairwise log-ratio change

$$\widehat{h}_t^*(x, i, i') := \log \frac{p_{t,x}^*(i)}{p_{t,x}(i)} - \log \frac{p_{t,x}^*(i')}{p_{t,x}(i')}. \quad (24)$$

For a neural policy π_θ , let

$$h_t(\pi_\theta; x, y, y') := \log \frac{\pi_\theta(y | x)}{\pi_t(y | x)} - \log \frac{\pi_\theta(y' | x)}{\pi_t(y' | x)}. \quad (25)$$

We project the finite-pool target into the LLM with

$$\mathcal{L}_{\text{reg}}^{(t)}(\theta) = \mathbb{E}_{x, (i, i') \sim \xi_x} \left[h_t(\pi_\theta; x, y_{x,i}, y_{x,i'}) - \widehat{h}_t^*(x, i, i') \right]^2. \quad (26)$$

No language-model partition function appears. At the inner optimum,

$$\widehat{h}_t^*(x, i, i') = \eta_t \sum_{k=1}^K \lambda_k \left(\widehat{Q}_{k,x}^*(i) - \widehat{Q}_{k,x}^*(i') \right), \quad (27)$$

which provides an independent solver and artifact check. Pairwise probabilities may come from direct human comparisons or a general pairwise preference model; NBPO never requires them to decompose into scalar response rewards.

Two projections of one solver target. Equation (26) constrains only *differences* of log-ratios inside the sampled support; it says nothing about the pool’s overall likelihood, and a network can drive its residual down while moving mass in ways the target never specified. We therefore also report a second projection of the same $p_{t,x}^*$, weighted behavior cloning over the pool,

$$\mathcal{L}_{\text{WBC}}^{(t)}(\theta) = \mathbb{E}_x \sum_{i=1}^{n_x} p_{t,x}^*(i) (-\log \pi_\theta(y_{x,i} | x)), \quad (28)$$

with the weights detached and $\log \pi_\theta$ the full-vocabulary autoregressive log-probability of the response, summed over its tokens and not divided by length. This is finite-pool weighted behavior cloning: an empirical approximation of the forward KL from the finite-pool target to the policy on the sampled support. It inherits none of the exactness of (24), and we do not present it as a realization of the population theorem. The two losses read the identical solver artifact and differ only in how that artifact is pushed into the network, which is what makes the comparison informative. Writing (28) on unordered pairs as $\frac{n_x}{2} (p_{t,x}^*(i) (-\log \pi_\theta(y_{x,i} | x)) + p_{t,x}^*(i') (-\log \pi_\theta(y_{x,i'} | x)))$ reproduces the prompt-wise weighted likelihood exactly when all $\binom{n_x}{2}$ pairs appear once, so both objectives share one dataloader, one tokenization and one token exposure.

The finite-pool solve is numerical and the LLM projection remains approximate. The convergence result below concerns exact population solutions of (16).

Algorithm 1 Nash Bargaining Preference Optimization

Require: Reference π_{ref} , pairwise models $P_{1:K}$, temperatures $\beta_{1:K}$, outer stages T .

- 1: Sample finite learner/comparator pools; construct pairwise and reference–reference tensors; estimate $\hat{d}_{1:K}$.
 - 2: Set $\pi_0 \leftarrow \pi_{\text{ref}}$.
 - 3: **for** $t = 0, \dots, T - 1$ **do**
 - 4: Score π_t on the learner pool and initialize positive λ .
 - 5: **repeat**
 - 6: Solve (23) independently for every prompt.
 - 7: Aggregate $\hat{s}_{1:K}$ and solve $\hat{s}_k = 1/\lambda_k$ for λ over Λ .
 - 8: **until** the projected dual/KKT criteria are met
 - 9: Construct $\hat{h}_t^*$ by (24) and verify (27).
 - 10: Fit one neural candidate $\tilde{\pi}_{t+1}$ using (26).
 - 11: Set $\pi_{t+1} \leftarrow \tilde{\pi}_{t+1}$ if held-out empirical surpluses are positive; otherwise retain π_t .
 - 12: **end for**
 - 13: **return** π_T .
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Table 1: Controlled nontransitivity on the circulant feasibility-preserving family, averaged over five independent controlled seeds (not policy or preference-model seeds). Values are $\min_k s_k/\rho^*$; higher is better. The three adaptive-game rows remain stable, while the matched fixed-reference and scalar-reward controls fail as BT deviance increases. [†]The legacy R -step solver converged on 0/25 cells and is not a valid performance arm. Full objective-wise results and residuals are in Appendix B.

α	BT dev.	NBPO	Game-KS	Game-util.	Fixed-ref.	Nash	BT-RM–Nash	BT-RM–util.	legacy R -step [†]
0.00	0.0003	0.881	0.893	0.866	0.883	0.883	0.883	0.867	0.840
0.25	0.0202	0.897	0.905	0.883	0.907	0.907	0.907	0.895	0.860
0.50	0.0813	0.914	0.921	0.900	0.809	0.809	0.809	0.800	0.082
0.75	0.1882	0.926	0.928	0.909	0.460	0.460	0.460	0.455	−1.511
1.00	0.3510	0.931	0.932	0.908	−0.241	−0.242	−0.242	−0.248	−3.373

Last-iterate convergence. Algorithm 1 returns the final outer policy rather than an average of past policies, so the relevant guarantee is on the last population iterate. Under Assumption 1, with $\pi_0 = \pi_{\text{ref}}$ and (16) solved exactly at every outer iteration with arbitrary $\eta_t > 0$, the Nash welfare is monotone, $F(\pi_{t+1}) \geq F(\pi_t)$, and $F(\pi^*) - F(\pi_T) \leq D(\pi^* \parallel \pi_{\text{ref}}) / \sum_{t < T} \eta_t$: a constant proximal step gives an $O(1/T)$ final-iterate gap. Theorem 1 and its proof are in Appendix A.6.

6 EXPERIMENTS

We separate the empirical questions the method induces. A feasibility-preserving controlled benchmark tests whether adaptive game values are needed as preferences become nontransitive; a human-labeled SAFERLHF panel tests whether Nash bargaining selects a useful helpfulness–harmlessness compromise; and neural-realization and general-capability evaluations test whether the finite-pool target transfers to an 8B policy. Separating them avoids reading sparse released comparison graphs as evidence of natural within-objective cycles.

6.1 CONTROLLED NONTRANSITIVITY ISOLATES PREFERENCE REPRESENTATION

Construction. Each instance contains 40 prompts and four objectives. Starting from a transitive payoff component $B_{k,x}$, we add an odd-cardinality circulant skew-symmetric component $C_{k,x}$ whose zero row sums leave the reference disagreement value unchanged. We use α as the nontransitivity knob: schematically, $A_{k,x}^{(\alpha)} = B_{k,x} + \gamma(\alpha)C_{k,x}$, where the calibrated scale $\gamma(\alpha)$ increases the cyclic pair interaction as α increases. A bisection calibrates this scale so that the maximum attainable worst surplus $\rho^* := \max_{\pi} \min_k s_k(\pi)$ remains approximately 0.03 while $\alpha \in \{0, .25, .5, .75, 1\}$ increases nontransitivity. No entrywise clipping is used. We run five independently generated seeds and evaluate every method with the same true adaptive-game values. The headline metric is $\min_k s_k(\pi)/\rho^*$, which normalizes the small remaining attainable-margin variation.

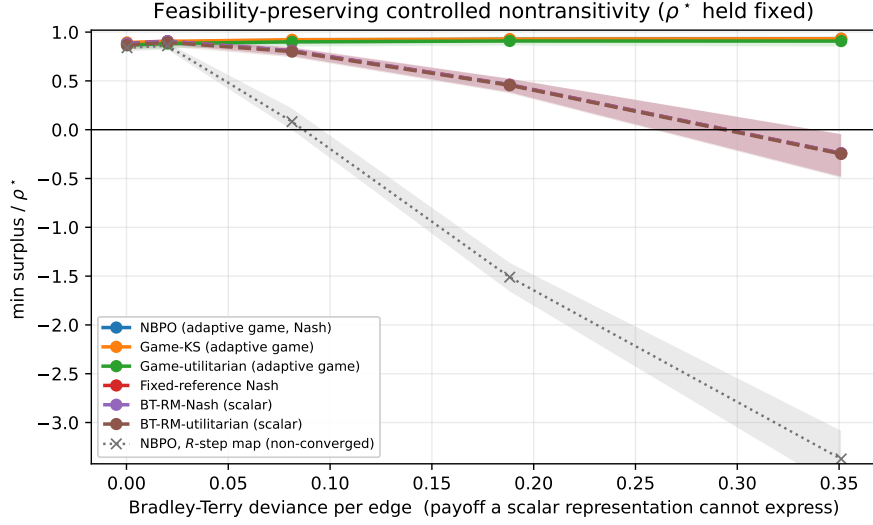


Figure 1: Feasibility-preserving controlled nontransitivity. Adaptive-game methods retain near-optimal worst-objective surplus as Bradley–Terry misspecification increases, whereas fixed-reference and scalar-reward representations deteriorate. The legacy R -step arm is shown only as a non-convergence ablation.

Representation, not an aggregation artifact. At $\alpha \leq .25$, all converged representations are similar. From $\alpha = .5$ onward, NBPO increasingly outperforms Fixed-reference Nash and BT-RM–Nash, reaching gaps of approximately 1.17 normalized units at $\alpha = 1$. BT-RM–Nash uses the same Nash rule as NBPO but collapses with its scalar-reward utilitarian counterpart, while NBPO, Game-KS, and game-utilitarian remain between 0.87 and 0.93. Thus the controlled result supports adaptive game-valued preference representation; it does not establish that Nash dominates every bargaining rule. Paired per-seed differences between NBPO and Game-KS do not reach significance at any α under a five-seed Student- t interval (for example -0.0007 , 95% CI $[-0.0347, +0.0332]$ at $\alpha = 1$); with five seeds this is an absence of evidence for a difference and not evidence of equivalence. The paired NBPO advantage over equal-weight game-utilitarian aggregation is positive in all 25 cells but reaches significance only at $\alpha = 0$ under the same interval.

Solver fidelity. On the controlled-v2 panel the direct prompt-wise solver meets every declared convergence gate in all 25 instances: projected KKT residual at most 6.2×10^{-9} , inner stationarity residual at most 5.1×10^{-5} against a 10^{-4} requirement — a factor of two of margin, not orders of magnitude — and finite-pool target identity at 1.5×10^{-14} . Table 6 decomposes exact, proximal and practical solutions on a *different* construction — the original interpolated family — where an exactly solved proximal reference is available; there the direct solve lies $TV = 0.027$ from that reference. Small inner residuals bound distance to the proximal solution at the current multipliers, not to a global Nash point. The legacy alternating map oscillates under stronger nontransitivity and fails every declared convergence gate. All final LLM experiments therefore require the direct finite-pool inner solve; Appendix E records the exact/proximal/practical decomposition and scaling measurements.

6.2 NATURAL OBJECTIVE CONFLICT ON SAFERLHF

SAFERLHF provides direct human pairwise labels for helpfulness and harmlessness (Dai et al., 2024). Its released graph contains no fully observed response triangles, so it cannot identify within-objective human cycles; we use it instead for the aggregation question. The two objectives disagree on 24.33% of held-out *human-annotated* pairs, which defines the conflict subset by a fixed prompt mapping. On *generated* response pools the frozen preference-model ensemble disagrees on only 7.7–7.8% of pairs (Table 10); the two rates have different populations, denominators and adjudicators and are not interchangeable. We fit matched-capacity objective models: an explicitly anti-symmetric joint-pair model for the general-preference interface and a scalar Bradley–Terry model for the reward-based

Table 2: Objective-wise surplus over the reference on 1,000 held-out SAFERLHF prompts, one training seed. Each policy is decoded greedily and scored against the *same* frozen finite comparator pools and the same disagreement point $\hat{d}$ used in training, so the rows differ only in the policy. $s_k = \hat{V}_k - \hat{d}_k$; intervals in the text are paired prompt bootstraps and exclude zero for both trained arms on both objectives. The finite-pool row is the solver’s own target p^* on the same prompts: the value the projection is aiming at, not a policy and not an upper bound — p^* is optimal over the *sampled pool*, while a policy may emit responses outside it and could in principle exceed that value. [†]Horizon selected on dev and declared before the arm was run. Greedy point-mass responses are not unbiased samples from the trained stochastic policy, so this table is a comparison of deployed decoders and not a certificate of Algorithm 1 acceptance; the run’s own evaluation record declines that certificate on the same grounds. Aggregation-rule controls — fixed-reference Nash, BT-RM–Nash, game-utilitarian, game-KS — and additional seeds are the next step and are not claimed here.

method	s_{help}	s_{harm}	$\min_k s_k$	med. tok.
finite-pool target p^*	+0.1653	+0.1662	+0.1653	—
Base = π_{ref}	−0.0405	−0.0320	−0.0405	19
NBPO-MSE, 1750	+0.1339	+0.1337	+0.1337	79
NBPO-WBC, 1750	+0.0738	+0.0808	+0.0738	48
NBPO-MSE, 250 [†]	—	—	—	—
NBPO-WBC, 250 [†]	—	—	—	—

controls. Their near-null held-out difference is reported rather than used as evidence for or against natural cyclicity. Final policy comparisons use the same sampled response pool and direct inner solver, and evaluation independent of the training preference models.

Table 2 reports what one seed of the repaired pipeline achieves on prompts held out from policy fitting. Both projections carry every objective strictly above the reference — the individual-rationality property the bargaining solution is chosen for — and the pairwise regression carries them furthest, to +0.134 and +0.134 against a base at −0.041 and −0.032, covering 85% of the distance from the base policy to the finite-pool target’s value (56% for weighted behavior cloning), a reference point rather than a bound. The two objectives land within 0.0002 of each other, which is the solution’s own signature rather than one objective subsidizing the other. Neither arm shortens its answers to get there: total generated tokens across the evaluation panel are 100% and 106% of the base model’s.

This panel cannot discriminate aggregation rules. It is worth saying before the comparison is asked for. The Nash dual assigns $\lambda = (6.325, 6.186)$ to helpfulness and harmlessness, 2.2% apart, so the two objectives are close to symmetric here. Equalizing those weights at the same ℓ_1 mass — the matched utilitarian control, which holds the effective KL strength fixed so that only the aggregation changes — moves the finite-pool target by a total-variation distance of 0.0008 on average and 0.0075 at worst, and changes the pairwise target by an RMS of 0.0034 against the target’s own RMS of 1.125. The two rules’ finite-pool surpluses agree to four decimals. Two policies fitted to targets that close would differ by far less than two seeds of the same target, so we do not run that arm and do not report a null from it: what the measurement shows is a property of this benchmark, not of aggregation. A discriminating test needs objectives whose Nash weights are genuinely asymmetric, and we treat that as a requirement on the next panel rather than a result about this one.

That does not settle the question, and two columns elsewhere say so. HARBENCH’s harmful-completion rate does not move for either arm (Table 17), so the improvement in the harmlessness *game value* does not appear on an external safety benchmark. And on general-purpose prompts an independent scalar reward model puts the weaker arm above the reference and the stronger arm clearly below it, reversing this table’s ordering; §6.3 takes that up.

6.3 NEURAL REALIZATION AND GENERAL CAPABILITY

The finite-pool solve and the neural fit are separate questions, and only the first is settled. We therefore report four columns that are easy to conflate: whether the solver artifact is self-consistent, whether the network fits the target on its own training pairs, whether that fit transfers to unseen prompts from the same pool construction, and whether the resulting policy is more useful. A regression threshold is a *diagnostic* on the second and third, never an acceptance criterion for the fourth: the earlier plan’s

Table 3: Held-out realization of the finite-pool target on the repaired pipeline. Diagnostics on 500 dev prompts, verified disjoint from the 2,000 training prompts; both projections read the identical solver artifact and differ only in how it is fitted. Each arm is shown at its own declared horizon; the full evaluation trajectories are in Table 15. [†] marks a horizon selected on dev and declared before the arm was run. nMSE is normalized by $\mathbb{E}[(\hat{h}_t^*)^2]$ on the same split, so 1.0 is the zero predictor. nMSE* is that residual at the single best rescaling of h_t , $1 - \mathbb{E}[h_t \hat{h}_t^*]^2 / (\mathbb{E}[h_t^2] \mathbb{E}[(\hat{h}_t^*)^2])$: the gap between the two columns is pure magnitude, so reading them together separates what the projection learned about the target’s *direction* from how far it then moved. The last column is the mean per-response log-ratio to the proximal centre, a drift measure rather than a fit measure. Every pre-repair arm in Table 14 sits at Pearson between -0.03 and -0.01 ; these numbers are what the eight corrections in Appendix E recover, and no single correction is credited with the recovery. *These are diagnostics. No checkpoint, horizon or hyperparameter in this paper is selected by them.*

Projection	Updates	nMSE	nMSE*	Sign acc.	Pearson	Spearman	$\overline{\log(\pi_\theta/\pi_t)}$
NBPO-MSE, 1750	1750	1.418	0.828	0.693	0.415	0.460	-2.07
NBPO-WBC, 1750	1750	13.070	0.977	0.486	-0.152	-0.105	-6.68
NBPO-WBC, 250 [†]	250	0.780	0.758	0.687	0.492	0.488	-0.51

held-out gate — normalized MSE below 0.90, sign agreement above 0.65, positive Pearson and Spearman — is reported as a measurement, and no checkpoint, horizon, or hyperparameter is selected by it. The floor of the normalized MSE is set by the target’s own estimation noise, so a small residual is not evidence that unseen pairs are predictable to the complement of it.

Table 3 reports both. The corrected pairwise regression transfers: at 250 updates it meets every pre-registered criterion at once — normalized MSE 0.839 against a 0.90 requirement, sign agreement 0.692 against 0.65, and positive Pearson and Spearman — with the direct-solver residual at 6.7×10^{-11} . Every arm trained before the corrections sat at Pearson between -0.03 and -0.01 with sign agreement 0.49–0.51 (Table 14), so this is a change of regime rather than of degree. It is also not attributable to any one correction: the repaired arms differ from those in target construction, reference handling, pool size and sampling temperature, and prompt count at once, and two earlier single-change arms were both negative.

At the pre-registered horizon the same arm meets three of the four criteria. Sign agreement is 0.693 and Spearman 0.460, neither of which ever leaves a narrow band over the whole run, but normalized MSE is 1.418: the induced log-ratio grows past the target’s own scale while the schedule is near its peak, and then contracts back toward it as the step size anneals (RMS 3.32 $\rightarrow$ 2.98 against a target RMS of 2.37, drift $-2.38 \rightarrow -2.07$ nats, both turning at 1000 updates). At the best single rescaling the residual is 0.828, so what the pre-registered gate rejects is magnitude, not direction. Weighted behavior cloning behaves differently in kind: it has no stationary point at the target, and it runs monotonically through zero correlation at 1000 updates into anti-correlation, with the pool’s mean log-likelihood still falling at the horizon. We therefore do not present it as an improvement on (26).

The two realizations are compared at a horizon fixed in advance: the canonical pairwise regression (26) and the weighted behavior cloning (28). They share the base model, the learner and comparator pools, the frozen preference-model ensemble, the solved p_t^* , the tokenization, the optimizer and the number of updates, so a difference between them isolates the projection rather than the bargaining objective. Final checkpoints are evaluated on task-specific and general benchmarks with no paid judge API and without the retired prompted-Qwen judge: in-domain SafeRLHF reward and cost models are labeled as diagnostics, a local reward model supplies a same-prompt reference win rate that is not an official leaderboard score, and instruction-following, mathematical and safety benchmarks use their official evaluators. Safety gains that arrive together with shorter answers and lower task accuracy are reported as a trade-off and not as a Pareto improvement.

Three evaluators, three orderings. They do not agree, and the disagreement is part of the result rather than an inconvenience in it. The training teacher puts the pairwise regression far ahead of weighted behavior cloning (+0.134 against +0.074 worst-objective surplus, Table 2). The official deterministic benchmarks put it at or slightly above the reference and put weighted behavior cloning

below (Table 17). An independent scalar reward model, scoring the same decoders on general assistant prompts, reverses that: same-prompt win rates against the reference of 0.419 [0.387, 0.452] and 0.578 [0.546, 0.612] on the ALPACAEVAL prompt set, both intervals excluding 0.5. It is not a length effect — the arm it penalizes writes *longer* answers than the reference — and additional training seeds would not remove a gap of that size. The arm that moves furthest toward a safety-and-helpfulness bargaining target is the one a general-purpose reward model likes least, off the distribution that target was defined on. We report that model as a third opinion rather than an adjudicator, since a scalar evaluator is exactly the object this formulation declines to assume; but no Pareto claim survives these three tables together, and we make none.

7 RELATED WORK

General pairwise preferences. NLHF and its scalable variants optimize directly against general pairwise preference models rather than fitting scalar rewards (Munos et al., 2024; Wu et al., 2025; Zhang et al., 2025; Calandriello et al., 2024; Rosset et al., 2024; Zhou et al., 2025). Blackwell-style methods extend this view to vector-valued feedback and target sets (Blackwell, 1956; Bhatia et al., 2020). PROSPER introduces a maximum-entropy Blackwell winner with a scalable single-policy update and an inner worst-case choice over objective weights and opponents (Zhang et al., 2026). NBPO instead keeps one regularized game value per objective and applies Nash bargaining across the improvements over a shared fallback.

Reward-based multi-objective alignment. MODPO, Rewarded Soups, and Panacea expose user-specified trade-offs through scalarization, policy interpolation, or preference conditioning (Zhou et al., 2024; Ramé et al., 2023; Zhong et al., 2024b). MaxMin-RLHF and RMOD optimize weakest-objective criteria, while MOPO designates a primary objective and lower-bound constraints for the others (Chakraborty et al., 2024; Son et al., 2026; Agnihotri et al., 2026). These methods are important empirical baselines because they explicitly fit or use scalar objective rewards, whereas NBPO learns from general pairwise feedback without assuming that each objective is transitive.

Social choice and bargaining. The Nash solution and its Pareto, uniqueness, and affine-invariance properties are classical (Nash, 1950). Social-choice work argues that alignment should state its aggregation rule explicitly (Conitzer et al., 2024). Multi-party RLHF studies utilitarian, Nash, and leximin welfare over party-specific learned rewards and separately considers reward-free von Neumann solutions (Zhong et al., 2024a); Nash bargaining has also been used to combine task gradients in multi-task learning (Navon et al., 2022). Our contribution is not a new bargaining axiom, but a formulation and practical optimizer for applying the classical Nash rule to policy-dependent objective-wise preference-game values.

8 CONCLUSION AND LIMITATIONS

NBPO separates two decisions in multi-objective alignment: adaptive game values represent general pairwise preferences within each objective, and Nash bargaining selects one reference-relative compromise across objectives. The constrained proximal formulation inherits the classical Nash outcome properties and admits an $O(1/T)$ exact-population last-iterate guarantee. Practically, a direct prompt-separable finite-pool inner solve is necessary: it reproduces the exact proximal target, whereas a naive alternating opponent-policy map can oscillate under strong nontransitivity. Feasibility-preserving experiments support the representation claim, with adaptive-game methods staying stable as scalar-reward and fixed-reference representations deteriorate; the natural SAFERLHF panel separately tests the aggregation claim. The evidence has clear limits. The controlled study is constructed, not a measurement of natural human cycles: the released datasets do not densely compare response triples, so they cannot establish within-objective intransitivity. More importantly, our 8B policies do not yet realize the finite-pool target on held-out prompts — the regression diagnostics in Appendix E do not meet the pre-registered criteria, so the LLM results are a report on that objective rather than a demonstration of it. Nash is also not empirically superior to Game-KS here, and Assumption 1 fails when no policy improves every objective; game values are averages, not per-prompt guarantees.

AI USE STATEMENT

In this work, generative AI tools were used to critique the problem formulation, assist with mathematical derivations and proof presentation, propose experimental checks, and edit manuscript text. The authors are responsible for independently verifying every theorem, citation, implementation, empirical result, and disclosure in the final submission.

ETHICS STATEMENT

This work studies aggregation rules for language-model alignment, including harmlessness as one objective. Improvement in a game value is not an absolute safety certificate. Safety-critical deployment should retain hard constraints and independent safety evaluations in addition to the bargaining objective.

REPRODUCIBILITY STATEMENT

Algorithm 1 distinguishes the population proximal problem, the numerically solved finite-pool inner problem, and its neural projection. Appendix A contains the dual and convergence proofs; Appendices B–G specify the controlled construction, natural supervision, hyperparameters, solver and neural diagnostics, and the prompted-judge audit. The submission package should include candidate pools, preference tensors, dual traces, solver residuals, per-seed outputs, and the code and configurations required to reproduce the reported results.

REFERENCES

- Akhil Agnihotri, Rahul Jain, Deepak Ramachandran, and Zheng Wen. Multi-objective preference optimization: Improving human alignment of generative models. In *International Conference on Machine Learning*, 2026.
- Kush Bhatia, Ashwin Pananjady, Peter L. Bartlett, Anca D. Dragan, and Martin J. Wainwright. Preference learning along multiple criteria: A game-theoretic perspective. In *Advances in Neural Information Processing Systems*, volume 33, pp. 7413–7424, 2020.
- David Blackwell. An analog of the minimax theorem for vector payoffs. *Pacific Journal of Mathematics*, 6(1):1–8, 1956.
- Ralph Allan Bradley and Milton E. Terry. Rank analysis of incomplete block designs: I. the method of paired comparisons. *Biometrika*, 39(3/4):324–345, 1952.
- Daniele Calandriello, Daniel Guo, Rémi Munos, Mark Rowland, Yunhao Tang, Bernardo Avila Pires, Pierre Harvey Richemond, Charline Le Lan, Michal Valko, Tianqi Liu, et al. Human alignment of large language models through online preference optimisation. *arXiv preprint arXiv:2403.08635*, 2024.
- Souradip Chakraborty, Jiahao Qiu, Hui Yuan, Alec Koppel, Furong Huang, Dinesh Manocha, Amrit Singh Bedi, and Mengdi Wang. Maxmin-rlhf: Alignment with diverse human preferences. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, 2024.
- Paul F. Christiano, Jan Leike, Tom Brown, Miljan Martic, Shane Legg, and Dario Amodei. Deep reinforcement learning from human preferences. In *Advances in Neural Information Processing Systems*, volume 30, 2017.
- Vincent Conitzer, Rachel Freedman, Jobst Heitzig, Wesley H. Holliday, Bob M. Jacobs, Nathan Lambert, Milan Mossé, Eric Pacuit, Stuart Russell, Hailey Schoelkopf, Emanuel Tewolde, and William S. Zwicker. Position: Social choice should guide AI alignment in dealing with diverse human feedback. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, 2024.
- Josef Dai, Xuehai Pan, Ruiyang Sun, Jiaming Ji, Xinbo Xu, Mickel Liu, Yizhou Wang, and Yaodong Yang. Safe RLHF: Safe reinforcement learning from human feedback. In *International Conference on Learning Representations*, 2024.

-
- John M. Danskin. *The Theory of Max-Min and Its Application to Weapons Allocation Problems*. Springer-Verlag, 1967.
- Rémi Munos, Michal Valko, Daniele Calandriello, Mohammad Gheshlaghi Azar, Mark Rowland, Daniel Guo, Yunhao Tang, Matthieu Geist, Thomas Mesnard, Andrea Michi, et al. Nash learning from human feedback. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, 2024.
- John F Nash. The bargaining problem. *Econometrica*, 18(2):155–162, 1950.
- Aviv Navon, Aviv Shamsian, Idan Achituve, Haggai Maron, Kenji Kawaguchi, Gal Chechik, and Ethan Fetaya. Multi-task learning as a bargaining game. In *Proceedings of the 39th International Conference on Machine Learning*, volume 162 of *Proceedings of Machine Learning Research*, 2022.
- Long Ouyang, Jeffrey Wu, Xu Jiang, Diogo Almeida, Carroll Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, John Schulman, Jacob Hilton, Fraser Kelton, Luke Miller, Maddie Simens, Amanda Askell, Peter Welinder, Paul Christiano, Jan Leike, and Ryan Lowe. Training language models to follow instructions with human feedback. In *Advances in Neural Information Processing Systems*, volume 35, pp. 27730–27744, 2022.
- Rafael Rafailov, Archit Sharma, Eric Mitchell, Stefano Ermon, Christopher D. Manning, and Chelsea Finn. Direct preference optimization: Your language model is secretly a reward model. In *Advances in Neural Information Processing Systems*, volume 36, 2023.
- Alexandre Ramé, Guillaume Couairon, Corentin Dancette, Jean-Baptiste Gaya, Mustafa Shukor, Laure Soulier, and Matthieu Cord. Rewarded soups: Towards pareto-optimal alignment by interpolating weights fine-tuned on diverse rewards. In *Advances in Neural Information Processing Systems*, volume 36, 2023.
- Corby Rosset, Ching-An Cheng, Arindam Mitra, Michael Santacrose, Ahmed Awadallah, and Tengyang Xie. Direct nash optimization: Teaching language models to self-improve with general preferences. *arXiv preprint arXiv:2404.03715*, 2024.
- Seongho Son, William Bankes, Sangwoong Yoon, Shyam Sundhar Ramesh, Xiaohang Tang, and Ilija Bogunovic. Robust multi-objective controlled decoding of large language models. In *International Conference on Learning Representations*, 2026.
- Amos Tversky. Intransitivity of preferences. *Psychological Review*, 76(1):31–48, 1969.
- Haoxiang Wang, Wei Xiong, Tengyang Xie, Han Zhao, and Tong Zhang. Interpretable preferences via multi-objective reward modeling and mixture-of-experts. In *Findings of the Association for Computational Linguistics: EMNLP 2024*, pp. 10582–10592, 2024.
- Yue Wu, Zhiqing Sun, Huizhuo Yuan, Kaixuan Ji, Yiming Yang, and Quanquan Gu. Self-play preference optimization for language model alignment. In *International Conference on Learning Representations*, 2025.
- Jiahao Zhang, Lujing Zhang, Keltin Grimes, Zhuohao Yu, Gokul Swamy, and Zhiwei Steven Wu. Back to blackwell: Closing the loop on intransitivity in multi-objective preference fine-tuning. *arXiv preprint arXiv:2602.19041*, 2026.
- Yuheng Zhang, Dian Yu, Baolin Peng, Linfeng Song, Ye Tian, Mingyue Huo, Nan Jiang, Haitao Mi, and Dong Yu. Iterative nash policy optimization: Aligning LLMs with general preferences via no-regret learning. In *The Thirteenth International Conference on Learning Representations*, 2025. URL <https://openreview.net/forum?id=Pujt3ADZgI>.
- Huiying Zhong, Zhun Deng, Weijie J. Su, Zhiwei Steven Wu, and Linjun Zhang. Provable multi-party reinforcement learning with diverse human feedback. *arXiv preprint arXiv:2403.05006*, 2024a.
- Yifan Zhong, Chengdong Ma, Xiaoyuan Zhang, Ziran Yang, Haojun Chen, Qingfu Zhang, Siyuan Qi, and Yaodong Yang. Panacea: Pareto alignment via preference adaptation for LLMs. In *Advances in Neural Information Processing Systems*, 2024b.

Runlong Zhou, Maryam Fazel, and Simon S. Du. Extragradient preference optimization (EGPO): Beyond last-iterate convergence for nash learning from human feedback. In *Conference on Language Modeling*, 2025.

Zhanhui Zhou, Jie Liu, Jing Shao, Xiangyu Yue, Chao Yang, Wanli Ouyang, and Yu Qiao. Beyond one-preference-fits-all alignment: Multi-objective direct preference optimization. In *Findings of the Association for Computational Linguistics: ACL 2024*, 2024.

A PROOFS

A.1 PROOF OF PROPOSITION 1

For a fixed comparator ν , define

$$\mathcal{J}_{k,\nu}(\pi) := J_k(\pi, \nu) + \beta_k \mathbb{E}_{x \sim \mathcal{D}} \text{KL}(\nu(\cdot | x) \| \pi_{\text{ref}}(\cdot | x)).$$

The first term is linear in π and the second does not depend on π , so $\mathcal{J}_{k,\nu}$ is affine. The pointwise minimum of affine functions is concave; hence V_{k,β_k} is concave. Since $|\Delta_k| \leq 1/2$,

$$|V_{k,\beta_k}(\pi) - V_{k,\beta_k}(\pi')| \leq \frac{1}{2} \mathbb{E}_{x \sim \mathcal{D}} \|\pi(\cdot | x) - \pi'(\cdot | x)\|_1,$$

which also proves continuity.

For $\beta_k > 0$, the inner minimization separates across prompts. Fix x and abbreviate $m(z) = m_k^\pi(z | x)$. The prompt-wise problem is

$$\min_{\nu \in \Delta(\mathcal{Y}_x)} \sum_z \nu(z) m(z) + \beta_k \sum_z \nu(z) \log \frac{\nu(z)}{\pi_{\text{ref}}(z | x)}.$$

The first-order condition under $\sum_z \nu(z) = 1$ gives

$$\nu(z) \propto \pi_{\text{ref}}(z | x) \exp\{-m(z)/\beta_k\},$$

which yields (8); substituting it back yields (9). The KL term makes the minimizer unique. Danskin’s theorem therefore gives, for every feasible direction h satisfying $\sum_y h(y | x) = 0$,

$$DV_{k,\beta_k}(\pi)[h] = \mathbb{E}_{x \sim \mathcal{D}} \sum_y h(y | x) \mathbb{E}_{z \sim \nu_{k,\pi}^*(\cdot | x)} \Delta_k(y, z | x).$$

Thus the coordinate derivative is

$$[\nabla_\pi V_{k,\beta_k}(\pi)]_{x,y} = \mathcal{D}(x) Q_k^\pi(y | x),$$

proving (10).

A.2 CONNECTION TO THE FIXED-REFERENCE

This subsection gives a finite-action illustration at the unregularized endpoint $\beta = 0$. It is not used in Assumption 1, the full-support analysis, or the finite-temperature experiments.

Define the fixed-reference improvement

$$a_k(\pi) = J_k(\pi, \pi_{\text{ref}}) = P_k(\pi \succ \pi_{\text{ref}}) - \frac{1}{2}. \quad (29)$$

Skew-symmetry gives $a_k(\pi_{\text{ref}}) = 0$.

Full cyclic example. For one prompt, let $\pi_{\text{ref}} = \delta_r$, where δ_y is the deterministic policy that always outputs y . Suppose A , B , and C each beat r with probability 0.75 and form the deterministic cycle

$$A \succ B, \quad B \succ C, \quad C \succ A.$$

Both

$$\pi_{\text{pure}} = \delta_A \quad \text{and} \quad \pi_{\text{mix}} = \frac{\delta_A + \delta_B + \delta_C}{3}$$

have the same fixed-reference improvement,

$$a(\pi_{\text{pure}}) = a(\pi_{\text{mix}}) = 0.75 - \frac{1}{2} = 0.25.$$

Their worst-opponent values are nevertheless different. For the pure policy, response C defeats A deterministically, so

$$V_0(\pi_{\text{pure}}) = \min_{\nu} J(\pi_{\text{pure}}, \nu) = J(\delta_A, \delta_C) = -\frac{1}{2}.$$

For the uniform mixture, consider first the pure opponent δ_A . The mixture outputs A , B , and C with equal probability. Against A , these responses respectively tie, lose, and win, giving centered payoffs 0, $-1/2$, and $+1/2$. Hence

$$J(\pi_{\text{mix}}, \delta_A) = \frac{1}{3} \left(0 - \frac{1}{2} + \frac{1}{2} \right) = 0.$$

By symmetry,

$$J(\pi_{\text{mix}}, \delta_B) = J(\pi_{\text{mix}}, \delta_C) = 0.$$

Against the reference,

$$J(\pi_{\text{mix}}, \delta_r) = 0.25,$$

because every response in the mixture beats r with probability 0.75. Since $J(\pi_{\text{mix}}, \nu)$ is linear in ν , no mixed opponent can achieve a value below the minimum over these pure opponents. Therefore

$$V_0(\pi_{\text{mix}}) = \min_{\nu} J(\pi_{\text{mix}}, \nu) = \min\{0, 0, 0, 0.25\} = 0.$$

Finally, the reference itself has

$$V_0(\pi_{\text{ref}}) = \min_{\nu} J(\delta_r, \nu) = -0.25,$$

since r loses to each of A , B , and C with probability 0.75. Thus

	$a(\pi)$	$V_0(\pi)$	$V_0(\pi) - V_0(\pi_{\text{ref}})$
π_{pure}	0.25	-0.50	-0.25
π_{mix}	0.25	0	0.25

The fixed reference therefore assigns the two policies the same score, even though the pure policy is fully exploitable by one response whereas the balanced mixture guarantees an average tie against every response in the cycle.

A.3 POLICY-INDUCED NASH BARGAINING AND CLASSICAL CONSEQUENCES

We prove Proposition 2. Since Π is compact and each V_{k, β_k} is continuous, $V_{\beta}(\pi)$ is bounded. The definition of $\mathcal{G}_{\beta}$ then implies boundedness. Closedness follows by taking any convergent sequence $v_n \in \mathcal{G}_{\beta}$, choosing policies π_n with $d \leq v_n \leq V_{\beta}(\pi_n)$, extracting a convergent policy subsequence, and applying continuity. Downward closure is immediate from the definition.

For convexity, take $v^1, v^2 \in \mathcal{G}_{\beta}$ with witnesses π_1, π_2 . Objective-wise concavity from Proposition 1 gives, for $\alpha \in [0, 1]$,

$$V_{\beta}(\alpha\pi_1 + (1 - \alpha)\pi_2) \geq \alpha V_{\beta}(\pi_1) + (1 - \alpha)V_{\beta}(\pi_2)$$

componentwise, so $\alpha v^1 + (1 - \alpha)v^2 \in \mathcal{G}_{\beta}$. Assumption 1 provides a point strictly above d , proving nondegeneracy.

For any $v \in \mathcal{G}_{\beta}$ with $v > d$, some policy π satisfies $v \leq V_{\beta}(\pi)$. Since the Nash objective is coordinatewise increasing,

$$\sum_k \log(v_k - d_k) \leq \sum_k \log(V_{k, \beta_k}(\pi) - d_k) = F(\pi).$$

Conversely, every $\pi \in \Pi_+$ induces a feasible value vector $V_{\beta}(\pi)$. Hence (12) realizes the classical Nash solution over $(\mathcal{G}_{\beta}, d)$.

The remaining statements are standard Nash-bargaining consequences, but we record the short arguments for completeness. Since the log objective tends to $-\infty$ when any surplus approaches zero, every optimizer lies strictly above d . Coordinatwise monotonicity implies Pareto optimality. Strict concavity of $\sum_k \log s_k$ over value vectors implies that all optimal policies induce the same surplus, hence the same game-value vector, although the policy itself can be nonunique.

Finally, under independent positive affine transformations

$$V'_k(\pi) = c_k V_k(\pi) + b_k, \quad d'_k = c_k d_k + b_k, \quad c_k > 0,$$

the transformed surplus is $s'_k(\pi) = c_k s_k(\pi)$ and

$$\sum_k \log s'_k(\pi) = \sum_k \log s_k(\pi) + \sum_k \log c_k.$$

The added term is independent of π , so the optimizer set is unchanged.

Table 4: Classical population guarantees of common aggregation rules under joint improvement. The Nash row follows from the standard bargaining solution; the other rows are supported by the finite counterexamples below.

rule	strict IR	Pareto optimal	unique value	affine invariant
positive-weight utilitarian	no	yes	no	no
absolute maxmin	no	no	no	no
surplus maxmin	yes	no	no	no
Nash bargaining	yes	yes	yes	yes

A.4 COUNTEREXAMPLES FOR ALTERNATIVE AGGREGATION RULES

The positive guarantees in Table 4 follow directly from the corresponding objectives. Because $w_k > 0$, the positive-weight utilitarian objective $\sum_k w_k V_k(\pi)$ is strictly increasing in every game-value coordinate, so none of its optimizers can be Pareto dominated. Under Assumption 1, surplus maxmin also attains a strictly positive minimum surplus because a jointly improving policy exists. The examples below establish the guarantees that can fail for the alternative rules.

All examples use three responses, so a policy is a probability vector $\pi = (\pi_1, \pi_2, \pi_3) \in \Delta_3$, and use the uniform reference $\pi_{\text{ref}} = (1/3, 1/3, 1/3)$. For objective k , Δ_k is the skew-symmetric pairwise payoff matrix: $(\Delta_k)_{ij}$ is the centered preference margin of response i against response j . At $\beta_k = 0$,

$$V_k(\pi) = \min_j (\pi^\top \Delta_k)_j, \quad d_k = V_k(\pi_{\text{ref}}), \quad s_k(\pi) = V_k(\pi) - d_k.$$

Thus $\pi^\top \Delta_k$ contains the expected payoff of π against each pure opponent, and $V_k(\pi)$ is its smallest coordinate.

Because each V_k is the minimum of linear functions, the optimizers below are obtained from small linear programs. For example, equal-weight utilitarian optimization can be written as

$$\max_{\pi, t_1, t_2} t_1 + t_2 \quad \text{s.t.} \quad t_k \leq (\pi^\top \Delta_k)_j \quad \forall k, j, \quad \pi \in \Delta_3.$$

Absolute maxmin uses one variable t with $t \leq (\pi^\top \Delta_k)_j$, while surplus maxmin uses $t + d_k \leq (\pi^\top \Delta_k)_j$ for every k, j .

Utilitarian failure of strict improvement. Let

$$\Delta_1 = \begin{pmatrix} 0 & -1/2 & -1/2 \\ 1/2 & 0 & -1/2 \\ 1/2 & 1/2 & 0 \end{pmatrix}, \quad \Delta_2 = \begin{pmatrix} 0 & -1/2 & 1/4 \\ 1/2 & 0 & -1/2 \\ -1/4 & 1/2 & 0 \end{pmatrix}.$$

For the uniform reference,

$$\pi_{\text{ref}}^\top \Delta_1 = (1/3, 0, -1/3), \quad \pi_{\text{ref}}^\top \Delta_2 = (1/12, 0, -1/12),$$

and hence

$$d = (-1/3, -1/12).$$

First, joint improvement is feasible. The policy

$$\bar{\pi} = (1/4, 1/4, 1/2)$$

is only a feasibility witness; it gives

$$\bar{\pi}^\top \Delta_1 = (3/8, 1/8, -1/4), \quad \bar{\pi}^\top \Delta_2 = (0, 1/8, -1/16).$$

Therefore

$$V(\bar{\pi}) = (-1/4, -1/16), \quad s(\bar{\pi}) = V(\bar{\pi}) - d = (1/12, 1/48) > 0.$$

In contrast, solving the equal-weight utilitarian problem $\max_{\pi} V_1(\pi) + V_2(\pi)$ gives

$$\pi_U = (0, 1/5, 4/5).$$

For this policy,

$$\pi_U^\top \Delta_1 = (1/2, 2/5, -1/10), \quad \pi_U^\top \Delta_2 = (-1/10, 2/5, -1/10),$$

so

$$V(\pi_U) = (-1/10, -1/10), \quad s(\pi_U) = (7/30, -1/60).$$

Thus joint improvement is possible, yet a utilitarian optimum falls below the fallback on objective 2.

Absolute-maxmin failure of strict improvement. Let

$$\Delta_1 = \begin{pmatrix} 0 & -1/2 & 1/4 \\ 1/2 & 0 & 1/2 \\ -1/4 & -1/2 & 0 \end{pmatrix}, \quad \Delta_2 = \begin{pmatrix} 0 & 1/4 & -1/2 \\ -1/4 & 0 & 1/2 \\ 1/2 & -1/2 & 0 \end{pmatrix}.$$

Again,

$$d = (-1/3, -1/12).$$

Joint improvement is feasible, as witnessed by

$$\bar{\pi} = (1/3, 7/15, 1/5),$$

for which

$$V(\bar{\pi}) = (-4/15, -1/60)$$

and therefore

$$s(\bar{\pi}) = (1/15, 1/15) > 0.$$

Absolute maxmin instead solves

$$\max_{\pi} \min\{V_1(\pi), V_2(\pi)\},$$

whose linear-program solution is

$$\pi_{AM} = (0, 4/5, 1/5).$$

For this policy,

$$\pi_{AM}^\top \Delta_1 = (7/20, -1/10, 2/5), \quad \pi_{AM}^\top \Delta_2 = (-1/10, -1/10, 2/5),$$

so

$$V(\pi_{AM}) = (-1/10, -1/10)$$

and

$$s(\pi_{AM}) = (7/30, -1/60).$$

Thus absolute maxmin maximizes the smallest *absolute* game value while allowing objective 2 to fall below its own fallback.

A Pareto-dominated surplus-maxmin optimum. Let

$$\Delta_1 = \begin{pmatrix} 0 & -1/2 & 1/2 \\ 1/2 & 0 & 1/2 \\ -1/2 & -1/2 & 0 \end{pmatrix}, \quad \Delta_2 = \begin{pmatrix} 0 & 0 & 1/2 \\ 0 & 0 & 1/2 \\ -1/2 & -1/2 & 0 \end{pmatrix}.$$

The uniform reference gives

$$d = (-1/3, -1/6).$$

For objective 2, no policy can achieve $V_2(\pi) > 0$, while every mixture supported on the first two responses achieves $V_2(\pi) = 0$. Consequently,

$$s_2(\pi) \leq 1/6$$

for every policy, so the optimal surplus-maxmin value cannot exceed $1/6$.

The two policies

$$\pi_c = (1/3, 2/3, 0), \quad \pi_e = (0, 1, 0)$$

satisfy

$$V(\pi_c) = (-1/6, 0), \quad s(\pi_c) = (1/6, 1/6),$$

and

$$V(\pi_e) = (0, 0), \quad s(\pi_e) = (1/3, 1/6).$$

Both therefore maximize

$$\min_k s_k(\pi)$$

at $1/6$. However,

$$s(\pi_e) > s(\pi_c)$$

in the first coordinate and is equal in the second, so π_e Pareto dominates π_c . Thus the surplus-maxmin optimizer set can contain Pareto-dominated policies.

A.5 DUAL DERIVATION AND REGRESSION FIXED POINT

For multipliers $\lambda_k \geq 0$, the Lagrangian of (16) is

$$\mathcal{L}_t(\pi, u, \lambda) = \sum_k \log u_k + \sum_k \lambda_k (s_k(\pi) - u_k) - \eta_t^{-1} D(\pi \| \pi_t).$$

For fixed $\lambda_k > 0$, maximizing $\log u_k - \lambda_k u_k$ gives $u_k = 1/\lambda_k$ and value $-\log \lambda_k - 1$; if $\lambda_k = 0$, the maximum is unbounded. Substitution yields (17), and the remaining maximization over π is (18). Uniqueness of that maximizer and Danskin's theorem yield (19).

Assumption 1 provides strict feasibility: choose $\bar{\pi}$ with $s_k(\bar{\pi}) > 0$ and $0 < \bar{u}_k < s_k(\bar{\pi})$. Slater's condition therefore gives strong duality. Stationarity in u_k gives

$$\lambda_{t+1,k} = \frac{1}{u_{t+1,k}} > 0,$$

and complementary slackness then forces $u_{t+1,k} = s_k(\pi_{t+1})$. This proves Proposition 3.

Dual step used as a fallback. Stationarity $\hat{s}_k(p_t^*(\lambda)) = 1/\lambda_k$ can also be approached by projected subgradient *descent* on the dual — the update below subtracts the dual gradient, and the dual of the concave inner problem is minimized over λ —

$$\lambda^{(m+1)} = \Pi_\Lambda \left[\lambda^{(m)} - \gamma_m \left(\hat{s} \left(p_t^*(\lambda^{(m)}) \right) - (\lambda^{(m)})^{-1} \right) \right], \quad (30)$$

where $\gamma_m > 0$ is the dual step size, Π_Λ is Euclidean projection onto $\Lambda = [\lambda_{\min}, \lambda_{\max}]^K$ with $0 < \lambda_{\min} < \lambda_{\max}$, and the inverse is elementwise. On the pools reported here this iteration stalls at a projected KKT residual near 10^{-2} after 3000 steps, whereas the root solve reaches 10^{-9} in tens of inner solves; no reported result uses (30).

What the projected certificate certifies. Both the root solve and the fallback act on the box $\Lambda = [\lambda_{\min}, \lambda_{\max}]^K$, so the residual we report is the projected KKT residual of the *box-constrained* dual. That residual certifies the original problem only when the box is inactive at the returned λ : on an active bound the projection can annihilate a genuine stationarity violation, and a small number then certifies nothing. We therefore report bound activity alongside every residual and treat a run with an active bound as uncertified rather than converged. In the runs reported here the multipliers are interior by a wide margin — on the SAFERLHF pool, $\Lambda = [10^{-3}, 10^3]^2$ and the solved $\lambda = (6.33, 6.19)$ sit at least two decades from either bound in $\log_{10} \lambda$ — and the stationarity residual recomputed independently from the returned policy is 6.7×10^{-11} .

Finite-pool optimality and neural projection. Fix λ and let $\pi_t^*(\lambda)$ solve (18). First-order optimality gives, for every prompt and response pair in the support,

$$h_t(\pi_t^*(\lambda); x, y, y') = \eta_t \sum_k \lambda_k \left(Q_k^{\pi_t^*(\lambda)}(y | x) - Q_k^{\pi_t^*(\lambda)}(y' | x) \right). \quad (31)$$

The direct finite-pool program (23) is the sampled analogue of this strictly concave problem. Its normalized optimizer determines all finite-pool log-probability differences, so (24) is identified without computing a language-model normalizer. Equation (27) is the corresponding finite-pool KKT identity and is checked numerically before an artifact is accepted. With a full-support pair proposal, a zero-regression-error minimizer of (26) matches every target log-probability difference; normalization therefore recovers the finite-pool target distribution on the sampled support. This statement concerns the regression target, not the finite-sample optimization error of the neural model.

A.6 LAST-ITERATE CONVERGENCE

Theorem 1 (Last-iterate convergence). *Under Assumption 1, let π^* maximize (12), set $\pi_0 = \pi_{\text{ref}}$, and solve (16) exactly at every outer iteration with arbitrary $\eta_t > 0$. Then $F(\pi_{t+1}) \geq F(\pi_t)$ for $t \geq 1$, and for every $T \geq 1$,*

$$F(\pi^*) - F(\pi_T) \leq \frac{D(\pi^* \| \pi_{\text{ref}})}{\sum_{t=0}^{T-1} \eta_t}. \quad (32)$$

Work throughout in the $\mathcal{D}$ -weighted geometry: pair functions of (x, y) by $\langle f, g \rangle_{\mathcal{D}} = \mathbb{E}_{x \sim \mathcal{D}} \sum_y f(x, y) g(x, y)$, and let h be the *prompt-wise* negative entropy $h(\pi)(x) = \sum_y \pi(y | x) \log \pi(y | x)$, whose gradient at (x, y) is $1 + \log \pi(y | x)$. Its Bregman divergence in that geometry is exactly $D(\pi \| \pi_t) = \mathbb{E}_{x \sim \mathcal{D}} \text{KL}(\pi(\cdot | x) \| \pi_t(\cdot | x))$. The prompt expectation enters once, through the pairing, and is not also carried inside h . Because the objective is increasing in u , the policy part of (16) is equivalently

$$\pi_{t+1} \in \arg \max_{\pi \in \Pi_+} \left\{ F(\pi) - \frac{1}{\eta_t} D(\pi \| \pi_t) \right\}. \quad (33)$$

The feasible set is nonempty by Assumption 1, and $F(\pi) \rightarrow -\infty$ as any surplus approaches zero. Thus the maximizer exists in Π_+ even when the center π_t is the reference $\pi_{\text{ref}} \notin \Pi_+$. Because π_t has full support and the remaining objective has finite directional derivatives at an interior surplus point, the inward derivative of $-D(\pi \| \pi_t)$ excludes zero-probability coordinates. Induction therefore gives a full-support, positive-surplus π_{t+1} .

Let g_{t+1} be a supergradient of F at π_{t+1} . First-order optimality of (33) and concavity of F imply, for every $\pi \in \Pi_+$,

$$F(\pi) - F(\pi_{t+1}) \leq \langle g_{t+1}, \pi - \pi_{t+1} \rangle_{\mathcal{D}} \quad (34)$$

$$\leq \frac{1}{\eta_t} \langle \nabla h(\pi_{t+1}) - \nabla h(\pi_t), \pi - \pi_{t+1} \rangle_{\mathcal{D}} \quad (35)$$

$$= \frac{1}{\eta_t} [D(\pi \| \pi_t) - D(\pi \| \pi_{t+1}) - D(\pi_{t+1} \| \pi_t)]. \quad (36)$$

For $t \geq 1$, π_t is feasible in the next subproblem. Comparing the optimum with π_t gives

$$F(\pi_{t+1}) - \frac{1}{\eta_t} D(\pi_{t+1} \| \pi_t) \geq F(\pi_t), \quad (37)$$

so the objective values are nondecreasing from π_1 onward.

Set $\pi = \pi^*$ in (36), multiply by η_t , and sum from $t = 0$ to $T - 1$:

$$\sum_{t=0}^{T-1} \eta_t (F(\pi^*) - F(\pi_{t+1})) \leq D(\pi^* \|\pi_{\text{ref}}). \quad (38)$$

Because the gaps are nonincreasing, the left-hand side is at least $(\sum_t \eta_t)(F(\pi^*) - F(\pi_T))$, which proves (32).

Corollary 1 (Convergence of the surplus profile). *Under the conditions of Theorem 1, let $s^* = s(\pi^*)$ denote the unique surplus vector induced by a Nash-optimal policy. Then*

$$\|s(\pi_T) - s^*\|_2^2 \leq \frac{2D(\pi^* \|\pi_{\text{ref}})}{\sum_{t=0}^{T-1} \eta_t}. \quad (39)$$

Hence a constant proximal step gives $\|s(\pi_T) - s^*\|_2 = O(T^{-1/2})$.

Proof. This is a stronger consequence of Nash-welfare convergence rather than a separate requirement for the NBPO update. For every objective, $V_k(\pi) \in [-1/2, 1/2]$, and $d_k = V_k(\pi_{\text{ref}}) \leq 0$ because $\nu = \pi_{\text{ref}}$ is feasible and $J_k(\pi_{\text{ref}}, \pi_{\text{ref}}) = 0$. Hence every positive surplus is at most one. On $(0, 1]^K$, the function $G(s) = \sum_k \log s_k$ is 1-strongly concave in Euclidean norm. Proposition 2 implies that all Nash-optimal policies induce the same game-value vector, and therefore the same surplus vector s^* . First-order optimality over the convex comprehensive surplus set induced by (14) gives

$$G(s^*) - G(s) \geq \frac{1}{2} \|s - s^*\|_2^2. \quad (40)$$

Applying this inequality to $s = s(\pi_T)$ and using (32) proves (39). $\square$

B CONTROLLED BENCHMARK: CONSTRUCTION AND FULL RESULTS

Feasibility-preserving construction. For every seed we form a transitive base matrix and an odd-cardinality circulant skew-symmetric perturbation. Zero row sums keep the uniform reference disagreement invariant, while bisection selects the cycle magnitude so that $\rho^* = \max_{\pi} \min_k s_k(\pi)$ stays close to a target margin. We reject invalid probabilities rather than clip matrix entries. The resulting panel increases BT deviance from approximately 2.7×10^{-4} to 0.351 while retaining positive joint improvement in every instance. All policies are evaluated under the same adaptive-game values, not their method-native scales.

Robustness across construction and temperature. The primary construction is checked against an independently generated projected skew-symmetric family, $C = N(R - R^T)N$ with N the symmetric projector annihilating both μ and a witness policy, and against a temperature sweep on that family. The trend replicates: adaptive-game rows stay flat while fixed-reference and scalar-reward rows fall. The effect *size* does not: the $\alpha = 1$ gap is +0.111 on the projected family against +1.172 on the circulant one, because the projected family’s BT deviance only reaches 0.174 against 0.351. We therefore claim robustness of the trend and do not attach the circulant effect size to it. Intervals in Table 7 are exact paired Student- t intervals over seeds, not bootstrap intervals.

C NATURAL SUPERVISION AND PREFERENCE MODELS

Dataset audit. Table 8 records the empirical role of each released dataset. ULTRAFEDBACK supplies complete per-completion ordinal attributes, so score-induced pairwise labels are transitive by construction. SAFERLHF supplies direct helpfulness and safety comparisons but almost exclusively isolated edges; zero observed triangles means the data cannot test within-objective cycles. It nevertheless contains substantial cross-objective disagreement and is therefore appropriate for the bargaining experiment.

Table 5: Full controlled-v2 results, averaged over five seeds. “Solver resid.” is the method-appropriate inner or KKT residual; an entry of zero for a non-adaptive method does not certify an adaptive fixed point. The legacy R -step arm is a non-convergence ablation.

method	min surplus	sd	min s/ρ^*	sd	exploitability	inner resid.
$\alpha = 0.00, \rho^* = 0.0300, BT\ dev. = 0.0003$						
NBPO direct	+0.0264	0.0011	0.881	0.037	0.0687	2×10^{-12}
Game-KS	+0.0268	0.0005	0.893	0.017	0.0685	10^{-9}
Game-util.	+0.0260	0.0014	0.866	0.048	0.0688	3×10^{-10}
Fixed-ref. Nash	+0.0265	0.0011	0.883	0.036	0.0688	0
BT-RM-Nash	+0.0265	0.0011	0.883	0.036	0.0688	0
BT-RM-util.	+0.0260	0.0014	0.867	0.047	0.0689	0
legacy R -step [†]	+0.0252	0.0012	0.840	0.041	0.0675	5×10^{-10}
$\alpha = 0.25, \rho^* = 0.0300, BT\ dev. = 0.0202$						
NBPO direct	+0.0269	0.0013	0.897	0.044	0.1091	10^{-7}
Game-KS	+0.0272	0.0007	0.905	0.025	0.1088	10^{-7}
Game-util.	+0.0265	0.0017	0.883	0.058	0.1094	4×10^{-8}
Fixed-ref. Nash	+0.0272	0.0012	0.907	0.040	0.1197	0
BT-RM-Nash	+0.0272	0.0012	0.907	0.040	0.1197	0
BT-RM-util.	+0.0269	0.0015	0.895	0.049	0.1198	0
legacy R -step [†]	+0.0258	0.0014	0.860	0.046	0.1063	7×10^{-1}
$\alpha = 0.50, \rho^* = 0.0300, BT\ dev. = 0.0813$						
NBPO direct	+0.0274	0.0013	0.914	0.044	0.1454	2×10^{-6}
Game-KS	+0.0276	0.0007	0.921	0.023	0.1450	2×10^{-7}
Game-util.	+0.0270	0.0017	0.900	0.057	0.1458	2×10^{-7}
Fixed-ref. Nash	+0.0243	0.0018	0.809	0.059	0.1945	0
BT-RM-Nash	+0.0243	0.0018	0.809	0.058	0.1945	0
BT-RM-util.	+0.0240	0.0019	0.800	0.064	0.1947	0
legacy R -step [†]	+0.0025	0.0044	0.082	0.147	0.1981	10×10^{-1}
$\alpha = 0.75, \rho^* = 0.0300, BT\ dev. = 0.1882$						
NBPO direct	+0.0278	0.0013	0.926	0.042	0.1693	8×10^{-6}
Game-KS	+0.0278	0.0005	0.928	0.018	0.1692	8×10^{-7}
Game-util.	+0.0273	0.0017	0.909	0.056	0.1696	7×10^{-7}
Fixed-ref. Nash	+0.0138	0.0027	0.460	0.089	0.2723	0
BT-RM-Nash	+0.0138	0.0027	0.460	0.089	0.2723	0
BT-RM-util.	+0.0137	0.0028	0.455	0.095	0.2724	0
legacy R -step [†]	-0.0453	0.0060	-1.511	0.201	0.2626	10×10^{-1}
$\alpha = 1.00, \rho^* = 0.0283, BT\ dev. = 0.3510$						
NBPO direct	+0.0265	0.0033	0.931	0.041	0.1824	5×10^{-5}
Game-KS	+0.0264	0.0027	0.932	0.019	0.1825	3×10^{-6}
Game-util.	+0.0258	0.0037	0.908	0.063	0.1826	2×10^{-6}
Fixed-ref. Nash	-0.0063	0.0068	-0.241	0.277	0.3464	0
BT-RM-Nash	-0.0063	0.0068	-0.242	0.276	0.3464	0
BT-RM-util.	-0.0065	0.0070	-0.248	0.284	0.3465	0
legacy R -step [†]	-0.0949	0.0064	-3.373	0.406	0.3337	10^0

Table 6: Finite-pool solver audit at $\alpha = 1$ in the original controlled family. The direct practical solve recovers the positive-surplus proximal target; the deployed legacy map does not.

solution	min surplus	$/\rho^*$	inner residual	TV to exact prox.
Exact global Nash	+0.0158	+0.935	–	0.007
Exact proximal Nash	+0.0158	+0.936	–	0.000
NBPO direct inner solve	+0.0139	+0.821	2×10^{-6}	0.027
Legacy R -step map	-0.1851	-10.938	10^0	0.643
BT-RM-Nash	-0.1001	-5.918	0	0.495

Anti-symmetric general preference model. For objective k , a joint encoder produces $a_{\phi,k}(x, y, z)$ and we set

$$\ell_{\phi,k}(x, y, z) = \frac{1}{2}\{a_{\phi,k}(x, y, z) - a_{\phi,k}(x, z, y)\}, \quad P_{\phi,k}(y \succ z \mid x) = \sigma(\ell_{\phi,k}(x, y, z)).$$

This guarantees exact swap complementarity and self-ties without constraining the logit to a scalar reward difference. In the SafeRLHF diagnostic, both the joint model and the scalar BT baseline use RoBERTa-base so that encoder capacity and training cost are matched; this backbone choice is an experimental control rather than part of the NBPO method. Because the observed SafeRLHF

Table 7: Robustness of the controlled trend. Gaps are the paired per-seed difference in $\min_k s_k / \rho^*$ at $\alpha = 1$, with an exact Student- t 95% interval over seeds (*not* a bootstrap interval). The circulant row and the projected skew-symmetric rows are different constructions; the β sweep was run on the projected family only. Converged counts exclude the legacy R -step ablation, which converges on no cell in any run.

cycle family	β	seeds	NBPO—FixedRef (t_{95})		NBPO—BT-RM (t_{95})		min ρ^*	converged
Circulant (primary)	0.25	5	+1.1724	[+0.8715, +1.4733]	+1.1733	[+0.8735, +1.4731]	0.0283	150/150
Projected skew-symmetric	0.25	5	+0.1114	[+0.0714, +0.1514]	+0.1115	[+0.0715, +0.1514]	0.0332	150/150
Projected skew-symmetric	0.10	3	+0.2381	[+0.0880, +0.3883]	+0.2385	[+0.0883, +0.3887]	0.0331	90/90
Projected skew-symmetric	0.50	3	+0.0448	[+0.0165, +0.0730]	+0.0449	[+0.0167, +0.0730]	0.0334	90/90

Table 8: Released-supervision audit. “Obs. triangles” counts fully observed closed triples of *human* comparisons, summed over both objectives and all three splits; zero means within-objective human cycles are unmeasurable here, which is not a measured cycle rate of zero. “Human conflict” is the released-label disagreement rate on the held-out split.

dataset	rows	prompts	responses	obs. triangles	human conflict	supervision	role
SAFERLHF	73,907	38,639	143,668	0	24.33%	direct pairwise	objective conflict
ULTRAFEEDBACK	63,967	63,967	4/prompt	n/a (score-induced)	n/a	ordinal attributes	transitive control

graph contains no triangles, their near-null comparison is a calibration result rather than evidence that human preferences are transitive.

D FINAL TRAINING PROTOCOL AND HYPERPARAMETERS

All method comparisons share prompts, candidate pools, preference tensors, tokenizer and sequence masks, policy-training budget, decoding, and evaluation. The core rows vary only the objective representation or aggregation rule: adaptive game versus fixed-reference margin versus scalar BT reward, and Nash versus utilitarian versus Kalai–Smorodinsky aggregation. Hyperparameters and operating points are selected on validation data before final seeds.

E SOLVER AND NEURAL-REALIZATION DIAGNOSTICS

The direct inner solve is exact only for the declared finite response pool and numerical tolerance. It does not make the preference model, finite-pool approximation, or neural projection exact. Final claims require both solver residuals and held-out regression diagnostics.

F EVALUATION PROTOCOL

Natural objective evaluation. The preferred headline evaluation is a blinded human comparison against the common reference, with helpfulness and harmlessness collected as separate tasks. Method identity is hidden, response order and opaque identifiers are randomized independently, ties are allowed, and individual votes are retained. Automatic Beaver reward/cost scores are reported only as in-domain diagnostics because those evaluators are trained from SafeRLHF-family supervision. HarmBench, IFEval, TruthfulQA, Arena-Hard, and AlpacaEval provide complementary safety and capability checks.

G PROMPTED-JUDGE AUDIT

A prompted Qwen3-32B route was audited before large-scale labeling and rejected. Increasing the decoding budget removed the original 4–21% truncation-induced invalid rate, but it did not repair semantic instability. Under the best deliberative protocol, confident swap consistency remained 0.412–0.708 against a predeclared 0.85 gate, and order split-half Spearman correlation remained 0.189–0.428 against 0.75. A factorial audit showed strong lexical A/B bias; opaque identifiers

Table 9: Held-out preference-model validation, seeds 41/42/43. All values come from the *checkpointed* training run whose weights are on disk and which every downstream scoring step uses; an earlier uncheckpointed run agreed to within ± 0.003 on every metric. Temperatures are fitted on validation only. The near-null GPM–BT difference is reported, not used as a gate: this dataset contains no observed triangles and cannot identify a cyclicity advantage.

model	objective	per-seed (mean of 3)			T	calibrated 3-seed ensemble				
		NLL	bal. acc	AUC		NLL	bal. acc	AUC	ECE	ens. sd
Anti-symmetric GPM	helpfulness	0.5543	0.7218	0.7999	1.282	0.5512	0.7257	0.8034	0.1228	0.0293
Anti-symmetric GPM	harmless.	0.5180	0.7556	0.8464	1.297	0.5131	0.7573	0.8512	0.1782	0.0357
Scalar BT	helpfulness	0.5560	0.7214	0.7974	1.270	0.5527	0.7250	0.8009	0.1220	0.0300
Scalar BT	harmless.	0.5212	0.7507	0.8431	1.265	0.5163	0.7544	0.8484	0.1779	0.0357

Table 10: Generated-pool geometry pilot on 200 SAFERLHF validation prompts. “Model conflict” is the frozen preference-model ensemble’s objective disagreement rate on *generated* responses and is not the human-label conflict rate of Table 8. “Pred. cycles” are *model-predicted* closed triangles on the comparator-versus-comparator tournament, where rows and columns index the same response set; no human triangle is observable in this data. Split-half correlations resample the oracle ensemble, not the pool. Selection used validation stability and compute only.

geom.	dup. rate	entropy	ens. sd	GPM/BT sign	model conflict	pred. cycles	split-half r	ρ	sign agr.	dual evals	solve (s)
4+4	0.0073	0.6117	0.0429	0.9102	7.67%	0/800	0.9172	0.9132	0.8667	42	24.8
8+8	0.0167	0.6120	0.0414	0.9061	7.80%	10/11200	0.9242	0.9180	0.8925	30	18.4

reduced that component but transferred comparable bias to physical position and still failed the gates. No prompted-judge bank, reward model, or policy checkpoint from these labels enters the main experiments.

Table 11: Final SafeRLHF policy-training configuration. Report the frozen values actually used by all final seeds.

setting	value	setting	value
Reference / learner	Llama-3.1-8B-Instruct	adaptation	full fine-tuning
Policy prompts train/dev/test	7,000/500/1,000	final seeds	42, 43, 44
Pool geometry	—	pair proposal	all unordered learner pairs
Preference models	3-seed GPM / BT ensembles	inner solver	direct finite-pool concave solve
Opponent temperature β	—	proximal coefficient η	—
Policy learning rate	—	optimizer updates	—
Optimizer / scheduler	Adafactor / cosine	warmup / weight decay	0.10/0
Global batch / max length	32/2048	precision / grad. clip	bf16 / 1.0
Dual tolerance / box	—	neural target mode	finite-pool log-ratio
Code / split / config hashes	—		

Table 12: Baseline contract. All matched controls use the same finite pool and neural realization.

method	within-objective representation	aggregation	changed component relative to NBPO
NBPO	adaptive regularized game	Nash	—
Fixed-reference Nash	fixed reference margin	Nash	removes adaptive opponent
BT-RM-Nash	scalar BT reward	Nash	imposes scalar reward representation
Game-utilitarian	adaptive regularized game	equal-weight sum	replaces Nash aggregation
Game-KS	adaptive regularized game	normalized proportional gains	alternative bargaining rule
BT-RM-utilitarian	scalar BT reward	equal-weight sum	scalar representation and aggregation control

Table 13: Neural realization of the finite-pool target. The zero-predictor MSE is $\mathbb{E}[(\hat{h}_t^*)^2]$ on the same split.

row	target RMS	p_{10}	p_{90}	$\ \lambda\ _1$	proj. KKT	inner resid.	identity	norm. MSE	sign agr.	Pearson
NBPO	2.273	−3.841	+1.092	13.71	4×10^{-15}	3×10^{-13}	2×10^{-15}	—	—	—
Fixed-ref. Nash	2.333	−3.959	+1.093	13.87	0	0	2×10^{-15}	—	—	—
BT-RM-Nash	2.473	−4.223	+1.059	2.88	0	0	2×10^{-15}	—	—	—
Game-util.	2.273	−3.843	+1.092	13.71	n/a	3×10^{-9}	2×10^{-15}	—	—	—
Game-KS	2.276	−3.848	+1.091	13.71	n/a	7×10^{-9}	2×10^{-15}	—	—	—
BT-RM-util.	2.476	−4.224	+1.059	2.88	n/a	0	2×10^{-15}	—	—	—

Table 14: Neural projection diagnostics for the arms trained *before* the implementation corrections of Appendix E; the repaired arms are in Table 3 and are not comparable row-by-row with these, which used sampled targets, a cached reference and a 4+4 pool warped to temperature 0.9. **No pre-repair arm satisfies the regression gate** (nMSE < 0.90 and sign agreement > 0.65, both on the selecting validation half, with Pearson and Spearman > 0). Three arms have *positive point estimates* of the frozen-pool worst-objective surplus $\min_k \mathbb{E}_x[s_k]$ on the test half; that is neither passage of the full neural gate nor completion of policy promotion, and no arm has positive surplus on validation. Columns state their split: nMSE is given for both halves, sign and the correlations are the test half, and surplus is given for both. π_t is the initial policy and p^* the finite-pool solver target; p^* performs no neural regression, so its gate is not applicable. Daggered arms are diagnostics, not frozen protocol candidates. Per-objective values, hashes and the evaluator version are in the result manifest.

arm	target	ref	upd.	N	nMSE		sign	Pearson	$\min_k \mathbb{E}_x[s_k]$		gate
					val	test			val	test	
π_t (start)	—	—	—	—	—	—	—	—	−0.0126	−0.0035	n/a
p^* (solver)	—	—	—	—	—	—	—	—	+0.1393	+0.1449	n/a
$\eta=0.1^\dagger$	sampled	cached	300	700	2.319	2.362	0.485	−0.011	−0.0137	−0.0038	fail
$\eta=0.3^\dagger$	sampled	cached	300	700	1.121	1.182	0.490	−0.030	−0.0106	−0.0036	fail
$\eta=1$	sampled	cached	300	700	1.020	1.026	0.505	−0.026	−0.0122	−0.0057	fail
lr 2e-7	sampled	cached	300	700	1.006	1.016	0.497	−0.008	−0.0087	−0.0026	fail
lr 1e-6	sampled	cached	300	700	1.020	1.028	0.512	+0.003	−0.0091	−0.0010	fail
lr 1e-5 †	sampled	cached	300	700	22.308	18.623	0.517	−0.008	−0.0066	−0.0067	fail
RB †	RB	cached	300	700	1.070	1.126	0.535	+0.025	−0.0087	−0.0023	fail
RB clip100 †	RB	cached	300	700	1.105	1.093	0.542	+0.064	−0.0130	−0.0001	fail
RB †	RB	cached	1200	700	1.074	1.077	0.560	+0.089	−0.0081	+0.0016	fail
canon †	canon	cached	1200	8	1.221	1.294	0.491	−0.025	−0.0126	−0.0036	fail
canon †	canon	cached	1200	50	1.253	1.229	0.522	+0.046	−0.0152	+0.0012	fail
canon †	canon	cached	1200	200	1.140	1.158	0.544	+0.050	−0.0097	−0.0016	fail
canon †	canon	cached	1200	700	1.088	1.090	0.570	+0.093	−0.0100	+0.0023	fail
canon †	canon	cached	300	700	1.072	1.112	0.545	+0.037	−0.0101	−0.0009	fail
canon †	canon	online	300	700	1.076	1.122	0.528	+0.013	−0.0098	−0.0025	fail

Table 15: Full held-out evaluation trajectories for the repaired arms, every recorded evaluation of every arm. Columns are as in Table 3. The two 1750-update arms are the pre-registered comparison; the two 250-update arms anneal their own cosine schedule over their own horizon and are not truncations of the long ones, which is why their drift is an order of magnitude smaller at the same learning rate.

Projection	Updates	nMSE	nMSE*	Sign acc.	Pearson	Spearman	$\log(\pi_\theta/\pi_t)$
NBPO-MSE, 1750	250	0.839	0.734	0.692	0.516	0.499	−1.13
	500	1.226	0.796	0.688	0.452	0.465	−1.86
	750	1.482	0.824	0.678	0.419	0.452	−2.42
	1000	1.568	0.839	0.694	0.401	0.449	−2.38
	1250	1.517	0.836	0.688	0.404	0.450	−2.27
	1500	1.432	0.831	0.691	0.412	0.455	−2.12
NBPO-WBC, 1750	1750	1.418	0.828	0.693	0.415	0.460	−2.07
	250	0.920	0.824	0.645	0.419	0.415	−0.75
	500	1.932	0.938	0.623	0.250	0.297	−2.04
	750	3.776	0.995	0.569	0.074	0.124	−3.27
	1000	6.121	0.999	0.535	−0.028	0.018	−4.39
	1250	11.611	0.979	0.496	−0.143	−0.096	−6.15
NBPO-WBC, 250 †	1500	12.672	0.979	0.489	−0.144	−0.097	−6.63
	1750	13.070	0.977	0.486	−0.152	−0.105	−6.68
	50	0.866	0.839	0.628	0.401	0.388	−0.39
	100	0.807	0.777	0.655	0.473	0.450	−0.55
	150	0.755	0.730	0.678	0.520	0.499	−0.64
	200	0.770	0.743	0.682	0.507	0.497	−0.57
	250	0.780	0.758	0.687	0.492	0.488	−0.51

Table 16: Direct finite-pool solver scaling. The 7,000-prompt row must be measured rather than extrapolated.

prompts	pool	workers	solve (s)	dual evals	s/eval	proj. KKT	inner resid.	peak RSS (MB)	artifact (s)
200	4+4	32	1.5	31	0.047	3×10^{-14}	3×10^{-11}	467	0.01
200	8+8	32	1.7	35	0.049	4×10^{-15}	5×10^{-12}	498	0.02
200	16+16	32	1.8	32	0.055	2×10^{-14}	10^{-11}	499	0.02
1000	4+4	96	4.4	33	0.132	7×10^{-15}	2×10^{-11}	499	0.03
1000	8+8	96	3.2	26	0.122	4×10^{-15}	6×10^{-12}	504	0.04
1000	16+16	96	4.4	37	0.119	3×10^{-13}	10×10^{-11}	534	0.07
7000	4+4	96	10.6	28	0.379	7×10^{-10}	4×10^{-11}	534	0.11
7000	8+8	96	10.9	27	0.404	2×10^{-15}	3×10^{-11}	544	0.22
7000	16+16	96	15.3	30	0.508	10^{-14}	7×10^{-11}	633	0.39

Table 17: General capability and safety transfer of the frozen SAFERLHF-trained checkpoints, under a protocol that calls no paid judge API. Official evaluators are used where they exist (IFEval’s own strict checker; GSM8K exact match under one fixed zero-shot chain-of-thought prompt and extractor, with the unparseable fraction reported beside it; the HarmBench classifier under its official generation truncation). “Ref. win” is a same-prompt comparison against the reference scored by one local reward model and is *not* an AlpacaEval or Arena-Hard leaderboard number; it is never named as one. Median response tokens are reported because a shorter answer moves every other column at once, and a safety gain that arrives with shorter, less accurate answers is a trade-off rather than a Pareto improvement. Em-dash: run not completed. The two short-horizon rows and their generation are still running; the reference-win columns invert the ordering the training teacher gives, which §6.3 discusses rather than reconciles.

method	IFEval strict	GSM8K EM	HarmBench ↓	Alpaca ref. win	Arena ref. win	med. tokens
Base = π_{ref}	0.752	0.867	0.281	0.500	0.500	457
NBPO-MSE, 1750	0.758	0.864	0.284	0.419	0.435	481
NBPO-WBC, 1750	0.741	0.852	0.256	0.578	0.594	457
NBPO-MSE, 250 [†]	—	—	—	—	—	—
NBPO-WBC, 250 [†]	—	—	—	—	—	—

[†]Horizon selected on the dev split from the 1750-update arms’ own regression trajectory, declared before these arms were run, and never presented as the pre-registered comparison.

Table 18: Blinded human-evaluation record. Fill from the released annotation manifest and imported labels.

item	final value
Prompts / methods / objectives	—
Annotators per pair / tie policy	—
Order and identifier randomization	independent; exact manifest hash —
Repeated quality-control fraction	—
Parse / completion rate	—
Inter-annotator agreement	—
Repeated-item consistency	—
Triangle coverage / observed cycle rate	—

Table 19: Summary of the retired prompted-judge route. Known-answer controls diagnose rubric understanding; natural-pair stability determines admissibility.

protocol	unresolved invalid	min control accuracy	natural-pair stability
P2-long, 3 templates	0.000	0.960	failed swap/order gates
P2-balanced	0.000	0.800	failed controls and stability
Opaque identifiers	0.000	0.940	failed swap/order gates